

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS:

Total Marks: 50

Annexure A

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Analytical Problem Solving

1. RSA Key Generation and Security Analysis

Given:

$p = 17, q = 19$

Compute n and $\phi(n)$

Choose $e = 5$ and calculate the private key d

Encrypt the message $M = 7$

Decrypt the ciphertext and verify correctness

Analyze the impact of choosing a small prime number on RSA security

2. ElGamal Encryption Analysis

Given:

$p = 23, g = 5, \text{ private key } x = 7, \text{ random } k = 3, \text{ message } M = 10$

Compute the public key

Perform encryption to generate ciphertext pair $(C1, C2)$

Decrypt the ciphertext to retrieve the original message

Analyze how randomness (k) improves security in ElGamal

3. RSA Digital Signature Verification

Given:

$p = 11, q = 17, e = 7, \text{ message } M = 8$

Compute n and $\phi(n)$

Find private key d

Generate the digital signature for the message

Verify the signature using the public key

Analyze how digital signatures ensure non-repudiation

4. Diffie-Hellman with Attack Resistance Evaluation

Given:

$p = 29, g = 2$

User A private key = 5

User B private key = 12

Compute public keys for both users

Compute the shared secret key

Analyze what happens if an attacker intercepts and replaces public keys

Suggest a cryptographic solution to ensure authenticity

5. Hybrid Encryption System Analysis

Given:

A system uses RSA for key exchange and AES for data encryption

RSA encryption time = 5 ms per key exchange

AES encryption time = 0.5 ms per data block

Total data blocks = 500

Calculate total time taken for encryption

Compare with using only RSA for encrypting all data

Analyze why hybrid encryption is more efficient

Justify its use in modern secure communication systems

1. RSA Key Generation and Security Analysis

Given:

- $p = 17$
- $q = 19$
- $e = 5$
- $M = 7$

Step 1: Compute n and $\phi(n)$

$$n = p \times q$$

$$= 17 \times 19$$

$$= 323$$

$$\phi(n) = (p - 1)(q - 1)$$

$$= 16 \times 18$$

$$= 288$$

Step 2: Find d

$$5d \equiv 1 \pmod{288}$$

$$d = 173$$

Public key: $(5, 323)$

Private key: $(173, 323)$

Step 3: Encryption

$$C = 7^5 \pmod{323}$$

$$C = 11$$

Step 4: Decryption

$$M = 11^{173} \pmod{323}$$

$$M = 7$$

Security Analysis

- Small primes are easy to factor.
 - Private key can be found quickly.
 - RSA becomes insecure.
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2. ElGamal Encryption Analysis

Given:

- $p = 23$
- $g = 5$
- $x = 7$
- $k = 3$
- $M = 10$

Public key

$$y = 5^7 \bmod 23$$

$$y = 17$$

$$\text{Public key} = (23, 5, 17)$$

Encryption

$$C_1 = 5^3 \bmod 23 = 10$$

$$C_2 = 10 \times 17^3 \bmod 23 = 2$$

$$\text{Ciphertext} = (10, 2)$$

Decryption

$$M = 2 \times 14^{-1} \bmod 23$$

$$M = 10$$

Randomness Analysis

- Produces different ciphertexts.
 - Prevents pattern attacks.
 - Improves security.
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3. RSA Digital Signature

Given:

- $p = 11$
- $q = 17$
- $e = 7$
- $M = 8$

Compute n and $\phi(n)$

$$n = 187$$

$$\phi(n) = 160$$

Find d

$$d = 23$$

Signature

$$S = 8^{23} \bmod 187$$

$$S = 173$$

Verification

$$173^7 \bmod 187 = 8$$

Signature verified.

Non-repudiation

- Confirms sender identity.
 - Ensures integrity.
 - Sender cannot deny the message.
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4. Diffie-Hellman

Given:

- $p = 29$
- $g = 2$
- $A = 5$
- $B = 12$

Public keys

$$A = 2^5 \bmod 29 = 3$$

$$B = 2^{12} \bmod 29 = 7$$

Shared key

$$K = 7^5 \bmod 29 = 16$$

Shared secret = 16

Attack analysis

If an attacker replaces public keys:

- Attacker gets both connections.
- Messages can be modified.

Solution

- Digital signatures
 - Certificates
 - PKI
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5. Hybrid Encryption

Given:

- RSA = 5 ms
- AES = 0.5 ms
- Blocks = 500

Hybrid time

$$= 5 + (500 \times 0.5)$$

$$= 255 \text{ ms}$$

RSA only

$$= 500 \times 5$$

$$= 2500 \text{ ms}$$

Analysis

- Hybrid is faster.
- RSA exchanges keys.
- AES encrypts data efficiently.
- Used in HTTPS and banking systems.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

